MTH345 Sample Midterm (100 pts + EC 20 pts)

Please print your name clearly. A penalty may be applied if your name is difficult to read.

Last Name: _____

First Name: _____

Instructions:

- There are nine problems. The points for each problem are indicated next to the question.
- To earn full credit, you must show all necessary work. If it is unclear which assumptions or theorems may be used without proof, write out as much of your reasoning as possible.
- This test is closed-notes, and no computing devices (including calculators) are allowed.

MTH345 Sample Midterm

P1 (10 pts)
Let A and $B_1, \ldots, B_n$ be events where the B_i are pairwise disjoint, $\bigcup_{i=1}^{n} B_i = S$, and $P(B_i) > 0$ for all i. Show that:

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}. \tag{1}$$

P2 (15 pts)
Let Z_X and Z_Y be the Z-scores of random variables X and Y, respectively.
a. (10 pts) Show that $\mathrm{Var}[Z_X + Z_Y] = 2(1 + \rho_{X,Y})$ and $\mathrm{Var}[Z_X - Z_Y] = 2(1 - \rho_{X,Y})$.

b. (5 pts) Hence, conclude that $-1 \leq \rho_{X,Y} \leq 1$.

P3 (15 pts)
Let $\boldsymbol{Z} = (Z_1, Z_2)^\top$ with $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \text{Unif}(-1, 1)$ be a latent vector, and let $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \epsilon_3)^\top$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ be observation noise independent of $\boldsymbol{Z}$. Suppose the observed vector $\boldsymbol{X} = (X_1, X_2, X_3)^\top$ is generated according to:

$$X_1 = Z_1 + \epsilon_1, \quad X_2 = Z_2 + \epsilon_2, \quad X_3 = (Z_1^2 - Z_2^2)/2 + \epsilon_3. \tag{2}$$

Express the marginal PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$.

P4 (10 pts)
Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \overset{\text{i.i.d.}}{\sim} \text{Multin}(1, \boldsymbol{\mu})$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_K)^\top$ lies on the probability simplex. Consider the Lagrangian function

$$\mathcal{L}(\boldsymbol{\mu}, \beta) = l_n(\boldsymbol{\mu}) - \beta \left(\sum_{k=1}^{K} \mu_k - 1 \right). \tag{3}$$

a. (5 pts) Find all stationary points of $\mathcal{L}(\boldsymbol{\mu}, \beta)$, and show that they satisfy $\mu_k \geq 0$ for all k.

b. (5 pts) Show that $l_n(\boldsymbol{\mu})$ is concave.

P5 (10 pts)
Let $\boldsymbol{X} \sim \text{Multin}(n, \boldsymbol{\mu})$. Calculate $\text{Var}[\boldsymbol{X}]$.

P6 (10 pts)
Suppose the first $k-1$ principal axes $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{k-1}$ have been determined as the orthonormal eigenvectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_{k-1}$ corresponding to the $(k-1)$-largest eigenvalues. Consider the Lagrangian function:

$$\mathcal{L}(\boldsymbol{w}_k, \beta, \boldsymbol{\gamma}) = \boldsymbol{w}_k^\top \hat{\Sigma} \boldsymbol{w}_k + \beta \left(1 - \boldsymbol{w}_k^\top \boldsymbol{w}_k\right) - 2 \sum_{j=1}^{k-1} \gamma_j \boldsymbol{w}_k^\top \boldsymbol{e}_j, \tag{4}$$

where $\hat{\Sigma}$ denotes the sample covariance matrix.

In this problem, you may use the spectral theorem without proof: any real symmetric positive semi-definite matrix A can be decomposed as $A = \sum_{j=1}^{d} \lambda_j \boldsymbol{e}_j \boldsymbol{e}_j^\top$, where the eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$, and the corresponding eigenvectors $\{\boldsymbol{e}_j\}_{j=1}^d$ form an orthonormal basis for $\mathbb{R}^d$.

a. (5 pts) Show that any stationary point $(\boldsymbol{w}_k, \beta, \boldsymbol{\gamma})$ of $\mathcal{L}(\boldsymbol{w}_k, \beta, \boldsymbol{\gamma})$ satisfies:

$$\hat{\Sigma}\boldsymbol{w}_k = \beta \boldsymbol{w}_k, \quad \|\boldsymbol{w}_k\|_2 = 1, \quad \text{and} \quad \boldsymbol{w}_k^\top \boldsymbol{e}_j = 0 \quad (\forall j < k). \tag{5}$$

b. (5 pts) Since the objective is continuous and $\{\boldsymbol{w} \in \mathbb{R}^d : \|\boldsymbol{w}\|_2 = 1,\ \boldsymbol{w}^\top \boldsymbol{e}_j = 0 \text{ for all } j < k\}$ is compact, a global maximizer exists and must be a stationary point. Show that:

$$\max_{\substack{\|\boldsymbol{w}_k\|_2 = 1 \\ \boldsymbol{w}_k^\top \boldsymbol{e}_j = 0 \ (\forall j < k)}} \boldsymbol{w}_k^\top \hat{\Sigma} \boldsymbol{w}_k = \lambda_k, \tag{6}$$

and that this maximum is achieved at $\boldsymbol{w}_k = \boldsymbol{e}_k$.

P7 (15 pts)
a. (5 pts) Verify that

$$f(t) = \frac{1}{2}\left(t \log t - (t+1) \log\left(\frac{t+1}{2}\right)\right) \tag{7}$$

satisfies $f(1) = 0$ and is strictly convex on $\mathbb{R}^+$.

b. (10 pts) Let $\mathcal{D}_{\mathrm{JS}}(\cdot \,\|\, \cdot)$ denote the Jensen-Shannon divergence. Show that

$$\mathcal{D}_{\mathrm{JS}}(f_1 \,\|\, f_2) = \int_{\mathbb{R}^d} f_2(\boldsymbol{x}) f\left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})}\right) d\boldsymbol{x}. \tag{8}$$

P8 (15 pts)
Let $(X_1, X_2, Y)^\top$ be generated from the following hierarchical model:

$$Y \sim \text{Ber}(p) \quad \text{and} \quad \boldsymbol{X} \mid Y = y \sim \mathcal{N}(\boldsymbol{\mu}_y, \sigma^2 \boldsymbol{I}_2), \tag{9}$$

where $0 < p < 1$, and $\boldsymbol{\mu}_0 \neq \boldsymbol{\mu}_1$.
a. (10 pts) Show that $\text{Cov}[X_1, X_2] = p(1-p)(\mu_{1,1} - \mu_{0,1})(\mu_{1,2} - \mu_{0,2})$.

a. (5 pts) Show analytically that $f_{X_1,X_2|Y}(x_1, x_2 \mid y) = f_{X_1|Y}(x_1 \mid y) f_{X_2|Y}(x_2 \mid y)$.

P9 (EC 20 pts)
In this problem, you may use the singular value decomposition (SVD) theorem without proof: any real matrix $A \in \mathbb{R}^{d_X \times d_Y}$ of rank r can be decomposed as

$$A = \sum_{j=1}^{r} \sigma_j \boldsymbol{p}_j \boldsymbol{q}_j^\top, \tag{10}$$

where the singular values satisfy $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$, and the left and right singular vectors $\{\boldsymbol{p}_j\}_{j=1}^{d_X}$ and $\{\boldsymbol{q}_j\}_{j=1}^{d_Y}$ form orthonormal bases for $\mathbb{R}^{d_X}$ and $\mathbb{R}^{d_Y}$, respectively.

a. (10 pts) Given centered data with sample cross-covariance matrix $S_{\boldsymbol{X},\boldsymbol{Y}} := (n-1)^{-1} \sum_{i=1}^{n} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{y}_i - \bar{\boldsymbol{y}})^\top \in \mathbb{R}^{d_X \times d_Y}$, the first pair of partial least squares (PLS) directions maximizes the sample covariance:

$$\begin{aligned} \max_{\boldsymbol{w}_1, \boldsymbol{v}_1} \quad & \boldsymbol{w}_1^\top S_{\boldsymbol{X},\boldsymbol{Y}} \boldsymbol{v}_1 \\ \text{s.t.} \quad & \boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1, \quad \boldsymbol{v}_1^\top \boldsymbol{v}_1 = 1. \end{aligned} \tag{11}$$

Consider the Lagrangian function:

$$\mathcal{L}(\boldsymbol{w}_1, \boldsymbol{v}_1, \beta_1, \beta_2) = \boldsymbol{w}_1^\top S_{\boldsymbol{X},\boldsymbol{Y}} \boldsymbol{v}_1 + \frac{\beta_1}{2} \left(1 - \boldsymbol{w}_1^\top \boldsymbol{w}_1\right) + \frac{\beta_2}{2} \left(1 - \boldsymbol{v}_1^\top \boldsymbol{v}_1\right). \tag{12}$$

Show that any stationary point $(\boldsymbol{w}_1, \boldsymbol{v}_1, \beta_1, \beta_2)$ of $\mathcal{L}$ satisfies:

$$S_{\boldsymbol{X},\boldsymbol{Y}} \boldsymbol{v}_1 = \beta \boldsymbol{w}_1, \quad S_{\boldsymbol{X},\boldsymbol{Y}}^\top \boldsymbol{w}_1 = \beta \boldsymbol{v}_1, \quad \|\boldsymbol{w}_1\|_2 = 1, \quad \text{and} \quad \|\boldsymbol{v}_1\|_2 = 1, \tag{13}$$

where $\beta := \beta_1 = \beta_2 = \boldsymbol{w}_1^\top S_{\boldsymbol{X},\boldsymbol{Y}} \boldsymbol{v}_1$.

b. (5 pts) Since the objective function is continuous on the compact constraint set $\{(\boldsymbol{w}_1, \boldsymbol{v}_1) : \|\boldsymbol{w}_1\|_2 = \|\boldsymbol{v}_1\|_2 = 1\}$, a global maximizer exists and is attained at a stationary point. Show that:

$$\max_{\substack{\|\boldsymbol{w}_1\|_2=1 \\ \|\boldsymbol{v}_1\|_2=1}} \boldsymbol{w}_1^\top S_{\boldsymbol{X},\boldsymbol{Y}} \boldsymbol{v}_1 = \sigma_1, \tag{14}$$

where σ_1 is the largest singular value of $S_{\boldsymbol{X},\boldsymbol{Y}}$, and that this maximum is attained at $(\boldsymbol{w}_1, \boldsymbol{v}_1) = (\boldsymbol{p}_1, \boldsymbol{q}_1)$.

c. (5 pts) Consider the following mirrored encoding-decoding mappings:

$$\begin{aligned} \boldsymbol{x} &\xrightarrow{\text{Embedding}} \boldsymbol{w}_1^\top(\boldsymbol{x}-\bar{\boldsymbol{x}}) \xrightarrow{\text{Reconstruction}} \bar{\boldsymbol{x}} + \boldsymbol{w}_1\boldsymbol{w}_1^\top(\boldsymbol{x}-\bar{\boldsymbol{x}}), \\ \boldsymbol{y} &\xrightarrow{\text{Embedding}} \boldsymbol{v}_1^\top(\boldsymbol{y}-\bar{\boldsymbol{y}}) \xrightarrow{\text{Reconstruction}} \bar{\boldsymbol{y}} + \boldsymbol{v}_1\boldsymbol{v}_1^\top(\boldsymbol{y}-\bar{\boldsymbol{y}}), \end{aligned} \tag{15}$$

where $\|\boldsymbol{w}_1\|_2 = \|\boldsymbol{v}_1\|_2 = 1$. Show that:

$$\begin{aligned} &\frac{1}{n-1}\sum_{i=1}^{n}\left\|\boldsymbol{x}_i - \left(\bar{\boldsymbol{x}} + \boldsymbol{w}_1\boldsymbol{w}_1^\top(\boldsymbol{x}_i-\bar{\boldsymbol{x}})\right)\right\|^2 + \frac{1}{n-1}\sum_{i=1}^{n}\left\|\boldsymbol{y}_i - \left(\bar{\boldsymbol{y}} + \boldsymbol{v}_1\boldsymbol{v}_1^\top(\boldsymbol{y}_i-\bar{\boldsymbol{y}})\right)\right\|^2 \\ &+ \frac{1}{n-1}\sum_{i=1}^{n}\left(\boldsymbol{w}_1^\top(\boldsymbol{x}_i-\bar{\boldsymbol{x}}) - \boldsymbol{v}_1^\top(\boldsymbol{y}_i-\bar{\boldsymbol{y}})\right)^2 \\ &= \operatorname{tr}(\hat{\Sigma}_{\boldsymbol{X}}) + \operatorname{tr}(\hat{\Sigma}_{\boldsymbol{Y}}) - 2\widehat{\operatorname{Cov}}[\boldsymbol{w}_1^\top\boldsymbol{X}, \boldsymbol{v}_1^\top\boldsymbol{Y}], \end{aligned} \tag{16}$$

where $\hat{\Sigma}_{\boldsymbol{X}}$ and $\hat{\Sigma}_{\boldsymbol{Y}}$ denote the sample covariance matrices of $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively.